A Guide to Qt Programming, and the Top Tools for the Task

Hard core Linux users prefer the power of the command line. In fact, in the early days, the command line was the sole user interface for Linux programs. As computing evolved, graphical user interfaces became an integral part of any computer application. Qt is a tool that facilitates the creation of GUIs.

Qt was primarily developed by Trolltech, a Norwegian software company that was acquired by Nokia in 2008. In August 2012, it was acquired from Nokia by Digia, a Finnish company. Qt is currently being developed both by the Qt Company, a subsidiary of Digia, and the Qt Project, under open source governance, involving individual developers and firms.

Qt is regarded as a free and open source widget toolkit for creating GUIs as well as cross-platform applications that run on various software and hardware platforms such as Linux, Windows, macOS, Android or embedded systems – with little or no change in the underlying code base, while still being a native application with native capabilities and speed. Qt allows developers to design their own program's user interface through a method that most end users are familiar with. Most of the GUI programs are created with Qt and they have a native looking interface, in which Qt is classified as a widget toolkit. Non-GUI programs such as command line tools and consoles for servers can also be developed.

Qt supports various compilers, including the GCC C++ compiler and the Visual Studio suite, apart from having extensive international support. Qt also provides Qt Quick, which includes a declarative scripting language called QML that allows the use of JavaScript to provide the logic. Rapid application development for mobile devices has become possible with Qt Quick, while logic can still be written with native code as well, in order to achieve the best possible performance. Other features include SQL database access, XML parsing, JSON parsing, thread management and network support.

Qt editions

Qt comes in several different editions, as mentioned below.

- ***Qt free edition:*** This is only available for UNIX/X11 and embedded systems. It contains all features of the Qt enterprise edition and can be only used to develop software that is licensed under GPL.
- ***Qt non-commercial edition:*** This is only available for MS Windows and contains all features of the Qt enterprise edition. It can only be used to develop non-commercial software.
- ***Qt professional edition:*** This is available for all supported platforms like UNIX/X11, embedded systems, MacOS and Windows. But it doesn't contain certain advanced features such as the table module, the XML module and the networking module.

Fundamental technologies in Qt

Qt is built on a set of core technologies provided by the QtCore and QtGui modules, and includes the following.

- *The Tulip Container Classes:* These are a set of template container classes.
- *The Arthur Paint System:* This is the Qt4 painting framework.
- *The Interview Framework:* This